

1. Overview

Goal: Analyzing the mutual interaction of different types of information, inspired from human perception which always combine different modalities:

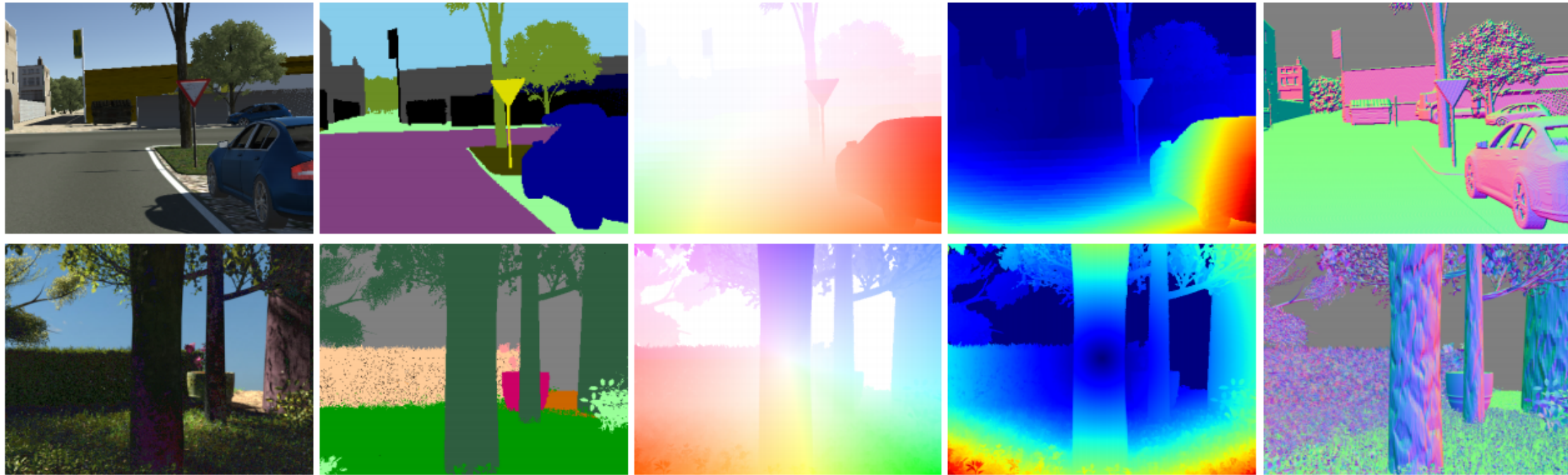
Objects' motion Optical flow

Objects' categories Semantic segmentation

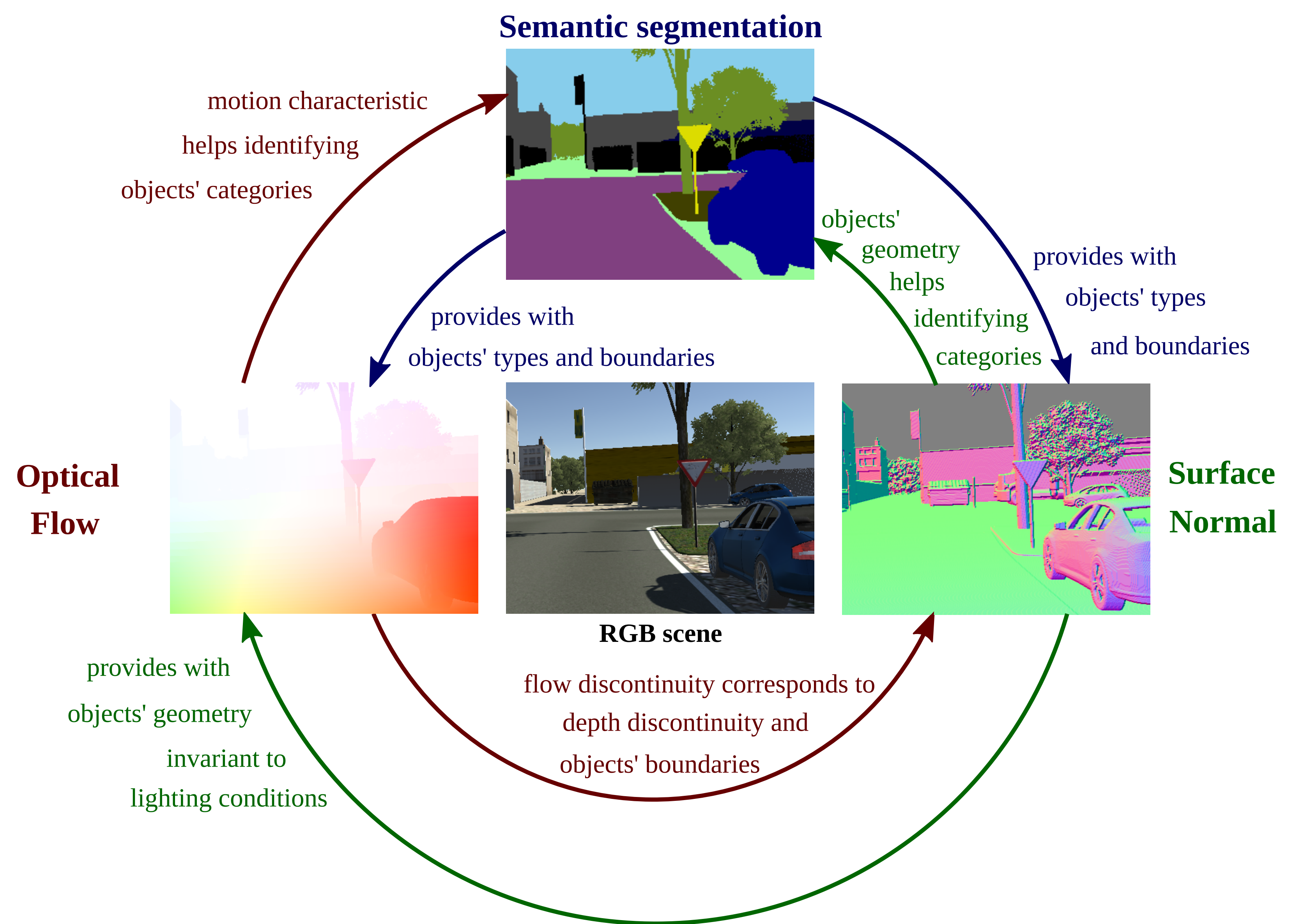
Objects' geometry Surface normals

Contributions:

- analysis the mutual interaction of 3 modalities
- combining 2 modalities to improve the other using CNN
- large scale synthetic dataset of outdoor nature scenes

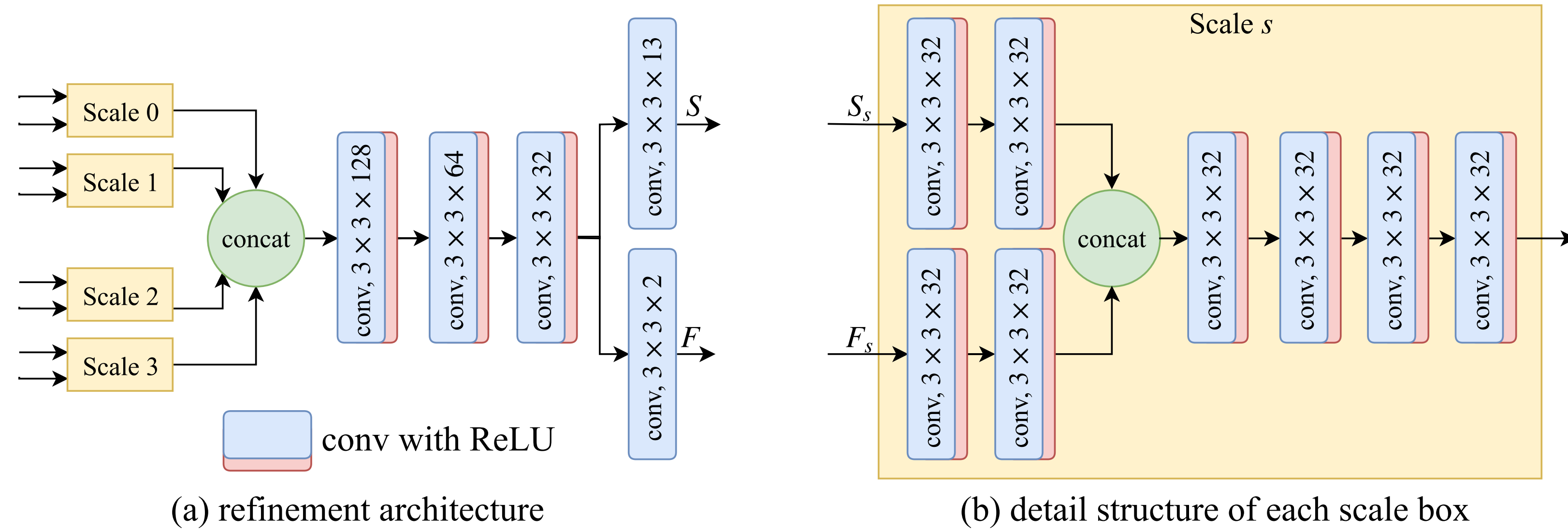


2. Motivation

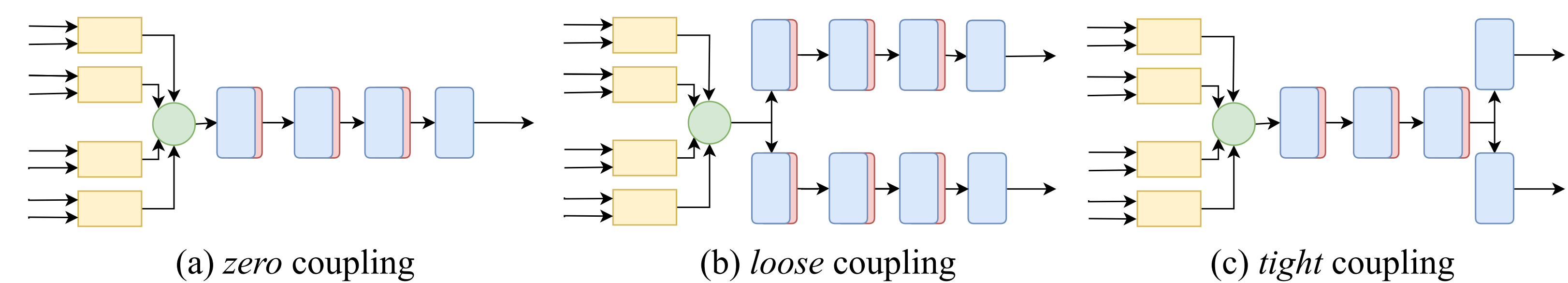


3. Method

Refinement architecture, inspired from [5]



Single task and multi-task refinement



4. Experiments

We refine each modality from the others (excluding RGB images) using either ground truth (GT) or predicted (PR) results to see how much extent one modality can benefit from others.

Datasets:

- virtual KITTI [4](20k images): synthetic street scenes
- Nature (15k images): synthetic nature scenes of gardens

Metrics

Optical Flow End-Point Errors (EPE) ↓

Segmentation Mean intersection-over-union (miou) ↑

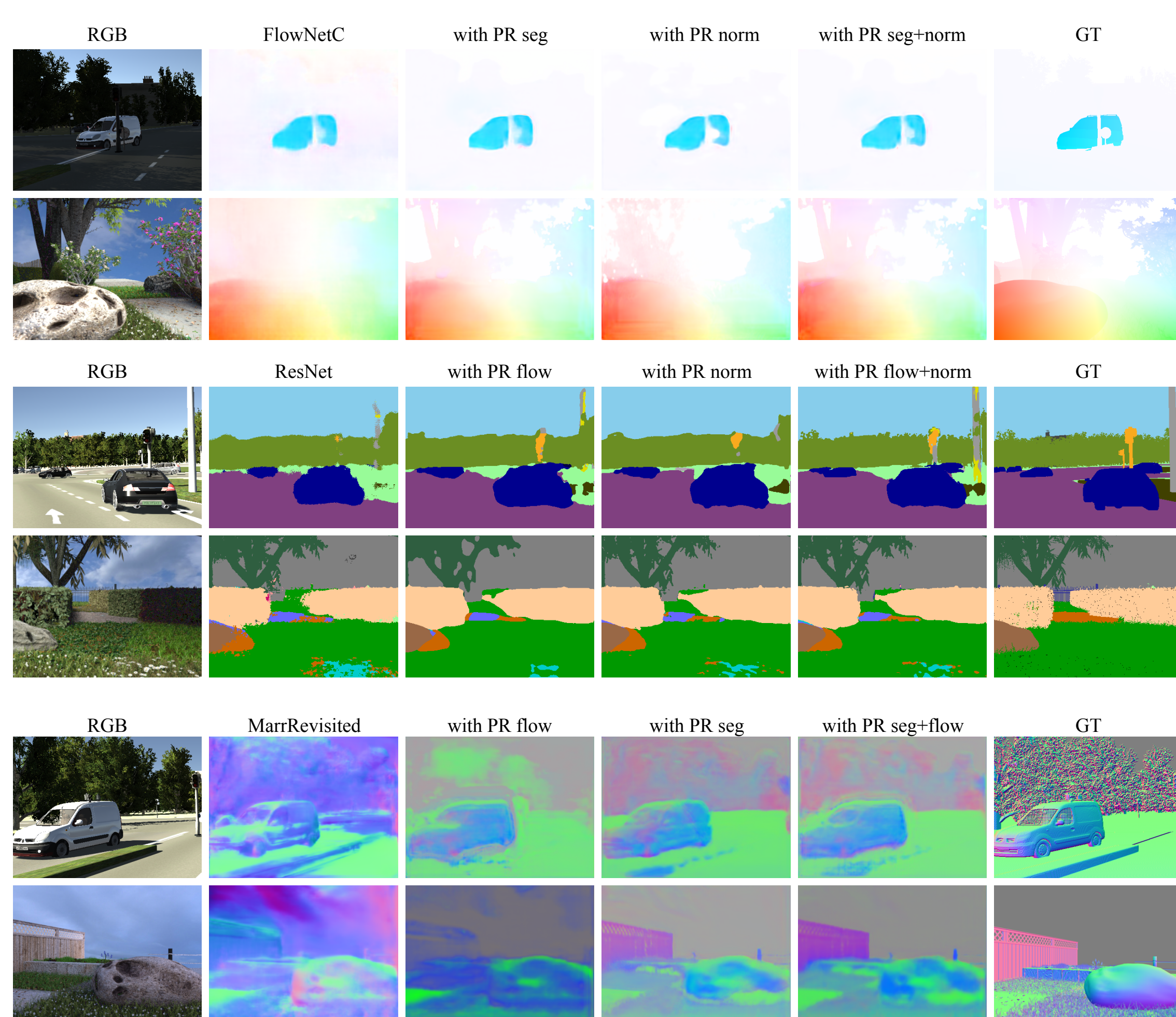
Surface normal mean ↓, median ↓, rmse ↓, 11.25 ↑, 22.5 ↑, 30.0 ↑

5. Refinement Couplings

Single task vs. multiple task refinement

Target	Baseline	GT zero	Predicted		
			zero	loose	tight+
Segmentation (miou↑)	44.11	46.9	44.78	41.2	41.1
Optical flow (epe↓)	2.68	2.37	2.40	2.43	2.41

6. Qualitative results



7. Quantitative results

Flow

Dataset	FlowNetC [3]	with GT seg	with GT norm	with PR seg	with PR norm	with PR seg+norm
VKITTI	2.68	2.37	2.36	2.40	2.50	2.39
Nature	16.19	14.09	13.92	14.16	16.62	14.21

Segmentation

Dataset	ResNet [2]	with GT flow	with GT norm	with PR flow	with PR norm	with PR flow+norm
VKITTI	44.11	46.90	50.0	44.78	45.36	47.55
Nature	37.88	38.4	41.6	37.57	38.83	38.00

Surface normals

		mean (↓)	median (↓)	rmse (↓)	11.25 (↑)	22.5 (↑)	30 (↑)
VKITTI	MarrRevisited [1]	48.13	48.34	57.44	17.39	19.81	27.44
	with GT flow	11.26	3.18	17.29	62.17	76.27	87.14
	with GT seg	11.16	3.17	16.78	60.70	75.67	88.43
	with PR flow	12.32	4.91	18.02	58.43	73.46	84.08
	with PR seg	11.44	2.82	17.24	60.30	74.02	86.58
Nature	MarrRevisited [1]	40.25	46.68	50.25	29.44	30.68	34.42
	with GT flow	9.62	8.01	13.20	61.78	89.74	97.53
	with GT seg	10.39	8.75	13.48	59.65	89.46	97.30
	with PR flow	11.39	9.80	14.38	55.36	86.88	96.66
	with PR seg	9.09	7.08	12.56	65.26	91.11	97.70

References

- [1] A. Bansal, B. Russell, and A. Gupta. Marr Revisited: 2D-3D Alignment via Surface Normal Prediction. In *CVPR*, 2016.
- [2] J. Cheng, Y.-h. Tsai, S. Wang, M.-H. Yang, and S. W. M.-h. Yang. SegFlow : Joint Learning for Video Object Segmentation and Optical Flow. In *ICCV*, 2017.
- [3] A. Dosovitskiy, P. Fischery, E. Ilg, P. Hausser, C. Hazirbas, V. Golkov, P. V. D. Smagt, D. Cremers, and T. Brox. FlowNet: Learning optical flow with convolutional networks. In *ICCV*, 2016.
- [4] A. Gaidon, Q. Wang, Y. Cabon, and E. Vig. Virtual Worlds as Proxy for Multi-Object Tracking Analysis. In *CVPR*, 2016.
- [5] O. H. Jafari, O. Groth, A. Kirillov, M. Y. Yang, and C. Rother. Analyzing modular CNN architectures for joint depth prediction and semantic segmentation. In *ICRA*, 2017.

Acknowledgement

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